

On this page

ETL Configurations

Ansible variables

Variable	type	Mandatory	Default Value	Description
stop_on_any_error	BOOLEAN		true	boolean flag to stop extraction if any error happened
num_extractions_threads	NUMBER		1	Number of threads assigned to extractions
num_publishing_threads	NUMBER		1	Number of threads assigned to publishing
max_events_per_second	NUMBER		5000	Max number of events that can be loaded per second
event_storage_splits	NUMBER		1	The number of event storage splits.
split_size	NUMBER		134217728	The size of the event storage splits.
papp_file_path	STRING		"{{ etl_home_path }}/var/tfrb.papp"	The path where the tfrb.papp is set
database_pulse_eventstorage_hostname	STRING	X		the host of the Events Storage Database
database_pulse_eventstorage_port	STRING		5432	The port of the Events Storage Database
database_pulse_eventstorage_schema	STRING		eventstorage	The schema of the Event Storage Database
database_pulse_eventstorage_username	STRING		"{{ lookup('gcp_secret_manager', secret='database_pulse_eventstorage_username') }}"	Pulse Event Storage username
database_pulse_eventstorage_password	STRING		"{{ lookup('gcp_secret_manager', secret='database_pulse_eventstorage_password') }}"	Pulse Event Storage password
storage_bucket_eventstorage_path	STRING	X		Path of the folder on bucket where the events will be stored, e.g: 'gs://BUCKET_NAME/FOLDER'
storage_bucket_eventstorage_folder_prefix	STRING		event_store_	Prefix of the bucket folder where the events will be stored, e.g: 'gs://BUCKET_NAME/FOLDER/event_store_'
cron_jobs	LIST[CRON]		daily extraction of last day	List of cron jobs to run
etl_extraction_interval	[INTERVAL]			Interval to be used by the extraction. Note: Only considered if <code>cron_jobs</code> is not defined or empty
etl_java_opts	STRING		"-Xms8g -Xmx8g"	

Variable	type	Mandatory	Default Value	Description
				ETL Java options. This property can also be used to override the metrics default port by setting it with <code>-Dcom.feedzai.commons.metmeknow.reporter.rest.DropwizardRestReporterFactory.port=<port></code>
pulse_connection_mode	STRING	X	"PAPP"	ETL Connection mode. <code>PAPP</code> indicates that a local application will be used whereas <code>RUNTIME</code> indicates that the application will be loaded from Pulse when the ETL process starts.
pulse_cluster_name	STRING			Mandatory for <code>RUNTIME</code> mode. Pulse cluster name in Zookeeper, usually set to <code>pulse{{ deployment_color }}</code>
zookeeper_nodes	STRING			Mandatory for <code>RUNTIME</code> mode. Zookeeper contact points (e.g. Zookeeper LB IP).
etl_include_result_tracking	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the result tracking should be exported or not
etl_include_formatted_result_tracking	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the formatted result tracking should be exported or not
etl_include_outcomes_extraction	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the outcomes should be exported or not
etl_outcomes_in_single_field	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the outcomes should be exported in a single field (JSON) or not (default - one column per outcome). Only works if <code>extraction.includeOutcome</code> or <code>extraction.includeFeedback</code> are enabled
etl_include_internal_state	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether or not internal state (scheduled metrics) will be included during extractions
etl_include_triggered_actions	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the triggered actions should be exported or not.
etl_project_all_fields	BOOLEAN		True if ETL configured for Historical Data Loading, true otherwise	The fields to be projected when <code>projectAllFields</code> is set to false. The order of the fields in this list will match the order in which they are projected to the targetAdapter.
etl_include_feedbacks	BOOLEAN			

Variable	type	Mandatory	Default Value	Description
			False if ETL configured for Historical Data Loading, false otherwise	Whether the exported events should include feedback or not. The default is false.
etl_include_workflow_state	BOOLEAN		False if ETL configured for Historical Data Loading, false otherwise	Whether the workflow state should be exported or not.
etl_include_event_diffs	BOOLEAN		False if ETL configured for Historical Data Loading, false otherwise	Whether the event diffs should be exported or not.
get_latest_published_app_version	BOOLEAN		false	Only applies to RUNTIME mode. If set to <code>true</code> , ETL will try to load the published workflow schema to extract events. If set to <code>false</code> (or if the current published schema is not found), ETL will load the current workflow schema.
etl_uuid_field	STRING		customer_id if ETL configured for Historical Data Loading, else use lifecycle_id	Indicates the CSV column where the event UUID is defined. When running ETL for Historical Data Loading, this property is set to <code>customer_id</code> to prevent inserting errors due to non-valid IDs.
etl_scheduled_metrics_manager	STRING		ALL_PLAN_EXECUTIONS	The type of the scheduled metrics manager to use when computing metrics. Can be one of <code>ALL_PLAN_EXECUTIONS</code> , <code>CURRENT_WORKFLOW</code> , <code>EMPTY</code> or <code>ZERO</code> .
use_rsync	BOOLEAN	X	true	Configures the deployment pipeline to use <code>rsync</code> instead of <code>copy</code> . Note that this variable should be set to <code>false</code> in RHEL8+ installations.
etl_cm_rabbitmq_queue_type	STRING		CLASSIC	Sets the RabbitMQ queue type (possible values are <code>CLASSIC</code> and <code>QUORUM</code>). Note that this impacts only Pulse to Case Manager recovery, and is ignored in normal data science extractions to GCS buckets.

Cron type

Variable	type	Mandatory	Default Value	Description
name	STRING	X		Name of the cron job
user	STRING	X		The user which will run the cron job
job_arguments	STRING	X		Arguments for the ETL tool Cron Job
hour	STRING		*	The cron expression for <code>hour</code>

Variable	type	Mandatory	Default Value	Description
minute	STRING		*	The cron expression for minute
day	STRING		*	The cron expression for day
weekday	STRING		*	The cron expression for weekday
month	STRING		*	The cron expression for month

Default cron jobs

```
cron_jobs:
- name: "Daily Extractions job of runtime data to Data Science environment"
  user: "{{ etl_unix_user }}"
  job_arguments: "--extraction -cf {{ etl_symlink_path }}/conf/event-storage-etl.properties -rei -ld --secret {{ etl_secret_path }}"
  hour: 0
  minute: 0
```

Copy

The `-rei` configuration follows [JTimeWindow Expressions](#).

In our example `-ld` means "the last day".

Other examples:

```
- Last 2 days: -2d/today
- Last week: -1w
- From 3 months ago until 1 month ago: -3M/-1M
- From 30 minutes ago until now: -30m/now
```

Copy

Interval type

Variable	type	Mandatory	Default Value	Description
start_event_timestamp	TIMESTAMP	X		Unix timestamp of the extraction interval start timestamp
end_event_timestamp	TIMESTAMP	X		Unix timestamp of the extraction interval end timestamp

Runtime mode notes

ETL Runtime mode enables ETL to fetch the current schema from Pulse before it starts exporting events, keeping the schema up to date without requiring a new release.

Please find below some notes regarding this mode.

Since ETL will connect through RMI to Zookeeper to fetch Pulse configuration and then Pulse to fetch the schema, the VM running ETL requires the following firewall rules to be set:

- TCP connection to Pulse port 1099 (this port is configured in Pulse property `pulse.manager.rmi.localregistryport`)
- TCP connection to Pulse port 1199 (this port is configured in Pulse property `pulse.manager.rmi.objectport`)
- TCP connection to Zookeeper port 2181

Important note

From 2.4.0 forward the below notice does no longer apply. By setting `get_latest_published_app_version` to `true` , ETL will extract events based on the published workflow schema, meaning that all columns the tool will try to fetch will already be present in the database.

ETL will fetch the schema even if the application is not published. The consequence of this is that the workflow schema can not have any changes because the database will not have the new column(s)

created. This consequence is not problematic since the schema is only changed when the API is also changed (with a new release). Since this process is done in a B/G deployment, ETL will always pick up the current schema.

Scheduled metrics manager🔗

ETL Runtime mode also supports schedule metrics extractions. This option can be enabled by setting `etl_include_internal_state` to `true`.

When extracting events from event storage to CSV a new configuration can be used to determine how ETL handles scheduled metrics extraction. In previous versions, when an event's scheduled metrics belonged to a plan that was no longer referenced in the workflow, extraction of the event would fail.

ETL now handles such cases gracefully and can also be configured to handle them in different ways. Find below the configuration options and how they influence schedule metrics extraction:

- `ALL_PLAN_EXECUTIONS` : ETL will search for the metrics definition in all existing plan executions
- `CURRENT_WORKFLOW` : ETL will search for metrics definitions in plan executions that are referenced in the workflow
- `EMPTY` : ETL will always return empty metrics
- `ZERO` : ETL will always return metrics with a value of zero